THE UNIVERSITY OF WESTERN AUSTRALIA

DESK No. | | | |

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Semester 2, 2024 EXAMINATIONS

School of Physics, Mathematics and Computing

CITS4403

Computational Modelling

Examination Duration: 2 hours

This is an examination WITH NO permitted materials

Provided by the University

1 x 18 Page Answer Booklet

Supplied by the Student

Calculator

No calculators are permitted

Instructions to Students

This exam paper is worth a total of 100 marks.

Your answers MUST be put in the provided Answer Booklet.

You need to clearly label which question you are answering.

Examination candidates may only bring authorised materials into the examination room. If you are found with unauthorised material, disciplinary action will be taken against you. This action may result in your being deprived of any credit for this examination or even, in some cases, for the whole unit. This will apply regardless of whether the material has been used at the time it is found. Any candidate who has brought unauthorised material into the examination room should declare it to the supervisor immediately. Candidates who are uncertain whether any material is authorised should ask the supervisor for clarification. Question papers and answer booklets must not be removed from the examination room.

Examination Cover Sheet

Part A: Graph (10 marks)

1. Describe the steps to construct a Watts-Strogatz Graph. **[4 marks]**

2. Compare how the Watts-Strogatz and Barabási-Albert models explain the 'small world' phenomenon in social networks. **[6 marks]**

Part B: Cellular Automata and Self-Organized Criticality (25 marks)

3. Explain the key differences between 1-dimensional and 2-dimensional Cellular Automata. **[6 marks]**

4. In a 2-D Cellular Automata that simulates percolation, each cell is either 'porous' with probability q or 'non-porous' with probability $1 - q$. When the simulation begins, all cells are considered 'dry' except the top row, which is 'wet'. During each time step, if a porous cell has at least one wet neighbour in the "von Neumann" neighbourhood, it becomes wet. Non-porous cells stay dry.

(a) How can one check if a randomly initialized array contains a percolating cluster? **[3 marks]**

(b) When estimating the probability of a percolating cluster, there is a critical value of q that indicates a rapid transition of the probability of a percolating cluster. Describe the process to estimate this critical value. **[6 marks]**

5. In a sandpile model with a critical value K=3, what are the possible values of the cells' states when the model reaches equilibrium. Explain your answers. **[4 marks]**

6. Visually identifying a fractal pattern of a 2D Cellular Automaton can be deceiving. Explain the detailed quantitative process for determining whether a pattern is fractal. **[6 marks]**

Part C: Agent-based Modelling (15 marks)

7. Explain the two main types of interaction in Agent-based Modelling? **[4 marks]**

8. Identify and explain three different instances where randomness is involved in the Sugarscape model. **[6 marks]**

9. How would you configure the Sugarscape model so that its simulation outcomes clearly exhibit emergent properties? Explain how it contributes to the emergence of complex behaviours. **[5 marks]**

Part D: Evolution & Evolution of Cooperation (35 marks)

10. A genotype of an agent is represented by a sequence of N binary digits. If N=4 and we start the simulation of evolution with one agent at every location of the fitness landscape, how many agents do we have before the simulation starts? Explain your answer. **[6 marks]**

11. Describe three different methods for initialising the locations of agents on a fitness landscape and justify the initial level of population variability for each method. **[9 marks]**

12. The following Python code simulates the process of population evolution. Identify which parts of the code you would modify to implement **differential survival** and **differential reproduction**. Describe in detail how you would change the code to introduce these two features and explain how the modified code would achieve these evolutionary dynamics. **[8 marks]**

```
class Simulation:

    def __init__(self, fit_land, agents):
        """Initialize with the fitness landscape and all the agents"""
        self.fit_land = fit_land
        self.agents = np.asarray(agents)

    def step(self):
        n = len(self.agents)
        fits = self.get_fitnesses()

        index_dead = self.choose_dead(fits)
        num_dead = len(index_dead)

        replacements = self.choose_replacements(num_dead, fits)
        self.agents[index_dead] = replacements

    def get_fitnesses(self):
        """Returns an array of agent fitnesses values."""
        fits = [agent.fitness for agent in self.agents]
        return np.array(fits)

    def choose_dead(self, ps):
        n = len(self.agents)
        is_dead = np.random.random(n) < 0.1
        index_dead = np.nonzero(is_dead)[0]
        return index_dead

    def choose_replacements(self, n, weights):
        agents = np.random.choice(self.agents, size=n, replace=True)
        replacements = [agent.copy() for agent in agents]
        return replacements
```

13. In the tournament of Prisoner's Dilemma, given a sequence of tuples [(None, None), (None, 'D'), (None, 'C'), ('C', 'D'), ('C', 'C'), ('D', 'D'), ('D', 'C')], where each tuple represents actions of the opponent in the previous two rounds. None means the rounds have not been played, 'C' means to cooperate, 'D' means to defect. To encode the tournament strategy as genotype, what is the genotype for (1) agents who always defect; (2) agents who always cooperate; (3) agents who apply the "tit for tat" (TFT) strategy. **[6 marks]**

14. Describe the process to evaluate the tendency of Niceness and Retaliating in the simulation of evolutionary dynamics of Prisoner's Dilemma tournaments. **[6 marks]**

Part E: Model Design Question (15 marks)

15. Design a Cellular Automaton (CA) to simulate the spread of a forest fire. Your model should include the following:
    (1) **Cell States**: Define the possible states of each cell. Explain what each state represents in the context of the forest fire.
    (2) **Initial Configuration**: Describe how the grid should be initialised. What are the starting conditions for the forest?
    (3) **Rules**: Specify the rules governing the fire spread across the forest.
    (4) **Randomization**: Explain how random factors are incorporated into the model.
    (5) **Quantifying the Outcome**: Propose a method to quantify the results of your simulation. What metrics will you use to assess the impact of the fire?

    [15 marks]

END OF EXAMINATION PAPER